

Encrypted Real-Time Communication Application Identification from Five Packet Measurements

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Abstract

Real-time communication (RTC) applications such as Discord, Google Meet, Messenger, WhatsApp and Zoom encrypt their media end-to-end with SRTP over DTLS, so payload inspection is unavailable. This work addresses the resulting identification problem: given only the *sizes and inter-arrival times of the first five packets* of an encrypted UDP media flow, predict the originating application and whether the call is voice or video—a ten-class problem. We contribute (i) a domain-grounded feature set of 241 deterministic descriptors derived from the encrypted-traffic-analysis literature; (ii) a leakage-aware grouped cross-validation protocol that accounts for the unpublished call identifiers in the dataset; (iii) a systematic three-round experimental campaign spanning 2,377 hyperparameter points, a self-supervised transformer pretrained on 90,610 auxiliary flows, tabular foundation models, and a set of customized linear probes; and (iv) a precise characterisation of the dataset’s *label geometry*, which shows that 160 of 1,285 training flows are provably unclassifiable because a video call whose first five packets contain no video fragment is physically indistinguishable from a voice call. The final architecture—a flat ten-class head averaged with a hierarchical application×mode composition, over a TabICL tabular foundation model, followed by one parameter-free decision rule—achieves **84.22% accuracy and 82.92% macro-F1 under grouped cross-validation**, against a measured ceiling of 93.77% accuracy and 93.64% macro-F1. We report all ablations, including the negative ones, and show that the residual error is concentrated in the irreducibly ambiguous Zoom voice/video pair.

Keywords: encrypted traffic classification; real-time communication; packet-length fingerprinting; tabular foundation models; grouped cross-validation; linear probes; self-supervised pretraining.

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1 Introduction

Real-time communication applications carry a substantial and growing fraction of interactive Internet traffic. Their media streams are protected end-to-end by SRTP over DTLS, which provides confidentiality, integrity and replay protection but also removes every avenue of deep-packet inspection. Identifying which application generated a flow—and whether a call is audio-only or audio-plus-video—must therefore proceed from *behavioural side-channels*: most notably the sizes and timings of encrypted packets.

The competition task formalises this as a ten-class classification problem over the cross product of five applications (Discord, Google Meet, Messenger, WhatsApp, Zoom) and two call modes (voice, video). The input for each flow is deliberately minimal: five packets, described by their cumulative arrival time and UDP payload length. The training set holds 1,285 labelled flows; the held-out test set holds 327.

This paper reports the full lifecycle of our solution. Three contributions stand out.

1. **A leakage-aware evaluation protocol.** The training flows are drawn from 400 source calls while the test flows come from 100 *held-out* calls, but no call identifier is published. Multiple flows from one call are near-identical, so a naive random split scores a memorised call rather than a generalising model. We introduce a pseudo-group key and evaluate everything under *grouped* cross-validation, which we argue is the only honest estimate available.
2. **A measured characterisation of the task’s ceiling.** We show that the dataset contains an exactly balanced, inseparable subset: 160 Zoom flows whose five-packet windows are audio-only, 80 of which are voice and 80 video, at coin-flip discriminability (grouped-CV AUC 0.55). This caps the entire task at 93.77% accuracy and 93.64% macro-F1, and it licenses one *parameter-free* decision rule that every model family we tested benefits from.
3. **A large, fully-reported ablation campaign.** Across three rounds we score 2,377 hyperparameter points, train a self-supervised transformer on a 6.97 GB auxiliary corpus, evaluate two tabular foundation models, build five linear probes, and test call-structure recovery—and we report the negative results with the same care as the positive ones, because they are what prevent a reader from repeating the same dead ends.

2 Related Work

Encrypted traffic classification rests on two observations. First, the codec and transport stack leaves a fingerprint in the *statistical distribution* of packet sizes: Taylor et al.’s *AppScanner* built classifiers from packet-size summaries; Draper-Gil et al.’s *ISCXVPN* added time-based flow features. Second, the *sequence* of sizes and inter-arrival gaps carries further structure: Korczyński & Duda used first-order Markov chains over quantised sizes; Shapira & Shavitt’s *FlowPic* treated the size–time plane as an image; Lopez-Martin et al. applied CNNs and RNNs directly to per-packet sequences.

Three strands from 2024–2025 are directly relevant to our later decisions. **Tabular foundation models** prior-fitted on synthetic datasets—TabPFN and its successors, TabICL—report state-of-the-art performance on datasets below 10,000 rows with no task-specific training, which is precisely our regime. **Traffic-specific pretrained transformers** (ET-BERT, YaTC, TrafficFormer, FlowletFormer) extend masked-modelling to packet streams. **Passive RTC measurement** (Michel et al.’s IMC’22 analysis of Zoom; recent work on encrypted video QoE) shows that even without payload, per-application header stacks and size families leak media type. Finally, the **MIRAGE** corpus from the University of Naples, and the MIMETIC/DISTILLER line of work built on it, supplies labelled mobile-application captures that we use as an auxiliary pretraining

and probe source.

Our evaluation protocol additionally engages with a methodological problem specific to traffic data: the *correlated-flows* property—several flows from one session—which, if ignored, inflates validation scores. Semi-supervised and clustering methods that exploit this property (X-Means with label propagation, SemTra) motivate one of our ablation branches.

3 Dataset and Problem Formulation

3.1 Data format

Each row is one media flow described by its first five packets, interleaved as $(\text{relative_time}_i, \text{packet_length}_i)$ for $i \in \{0, \dots, 4\}$. `relative_time_0` is identically zero; times are cumulative with microsecond precision; lengths are UDP payload sizes (total length minus the 8-byte UDP header). The target is one of ten case-sensitive strings encoding `application_mode`.

Table 1: Dataset summary.

Property	Value
Training flows / test flows	1,285 / 327
Columns per flow	10 + label
Packet length range	26–1,242 bytes
Five-packet time span	48 μ s–23.2 s (median 571 μ s)
Class imbalance	6.4 $\times$ (Discord_video 256 $\rightarrow$ GoogleMeet_voice 40)
Missing values / reordered timestamps	none

3.2 Integrity and leakage structure

Exploratory analysis established three facts that shaped everything downstream.

1. **The label is almost determined by the length tuple.** 1,136 distinct length tuples exist; only **2** map to more than one class, covering 95 rows (7.4%)—the repeated constant-size keepalive patterns. The Bayes floor is therefore near zero *outside* those tuples; the difficulty is generalisation, not label noise.
2. **Train and test are exchangeable.** A train-vs-test discriminator scores **AUC 0.529**, essentially chance, so cross-validation is a meaningful proxy for the leaderboard.
3. **Group structure is real but moderate.** 1,107 pseudo-groups; 253 flows (20%) share a near-duplicate signature with another flow. Because the test split is drawn from held-out calls, a plain stratified split leaks.

3.3 Evaluation protocol

Every model is scored under **two** schemes in parallel.

- **plain** — `RepeatedStratifiedKFold` (5 folds, 3 repeats, seed 42). The optimistic estimate, retained for comparability with published baselines.
- **group** — `StratifiedGroupKFold` (5 folds) over the pseudo-group key, repeated twice with different shuffles. **This is the decisive number** and every selection decision in this paper uses it.



Figure 1: The leakage mechanism introduced by unpublished call identifiers, and the grouped cross-validation protocol we adopt in response.

Fold indices are computed once and stored, so scores from different runs and containers are directly comparable. The primary metric is accuracy (implied by the submission format); macro-F1 and macro-recall are tracked as secondary metrics because the test distribution is explicitly not guaranteed uniform.

4 Methodology

4.1 System overview

4.2 Feature engineering

We engineer twelve feature blocks grounded in the encrypted-traffic literature. The decisive design choice is the split between *deterministic* blocks and *fitted* blocks (Figure 3).

The protocol-semantics block (F5) encodes what survives encryption: the four length bands (< 100 B STUN/RTCP/DTX, 100–300 B Opus+SRTP, 300–1100 B, 1100–1600 B path-MTU video fragments), padding residues (mod 4/8/16), and RTP-header-adjusted payloads (−12 B, −22 B for the SRTP authentication tag). The burst block (F6) segments each flow at four inter-arrival thresholds, because a video frame fragmented across the MTU arrives as one sub-millisecond group while audio at 20 ms ptime arrives as singletons.

The fitted blocks (F9–F11)—class-conditional Markov chains, Gaussian mixtures, and per-class neighbour distances—are computed with an *inner* out-of-fold loop so that training-side values are never in-sample. Despite this, Section 5 shows they cost ≈ 1.8 accuracy points and are disabled.

4.3 Model families

Thirteen configurations across twelve families, spanning deliberately different inductive biases (Table 2).

Table 2: Model families and their roles.

Family	Type	Notes
XGBoost / LightGBM / CatBoost / HistGB	boosted trees	<code>multi:softprob</code> / <code>multiclass</code>
RandomForest / ExtraTrees	bagged trees	<code>class_weight</code> searched
RBF-SVM / k-NN	kernel / distance	probability-calibrated
Logistic regression / MLP	linear / neural	standardised
PacketFormer	5-token transformer	Section 4.5
TabICL / TabPFN	tabular foundation models	prior-fitted, no task training

Class imbalance is treated as a *searched hyperparameter*—`{none, balanced, balanced_subsample}`—never as an assumption.

4.4 Distributed hyperparameter search

The search is materialised as an explicit list of parameter specifications, one container per point, which is a distributed `GridSearchCV` whose parallelism lives at the infrastructure layer. A second stage refines the grid winner with Optuna’s TPE sampler distributed by ask/tell batching, journalled so an interrupted driver resumes. Every point writes its own artifact, which makes the record reconstructible and crash-tolerant (Figure 4).

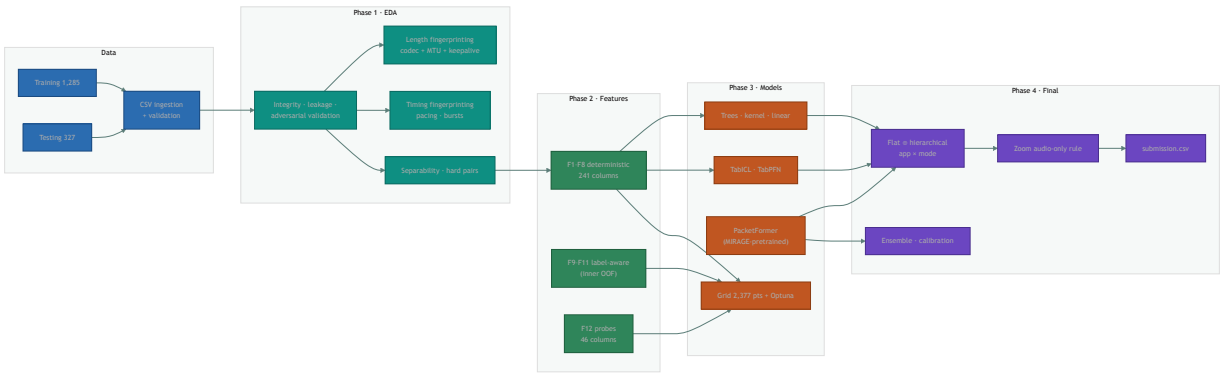


Figure 2: End-to-end pipeline, from ingestion through the four phases to the final submission.

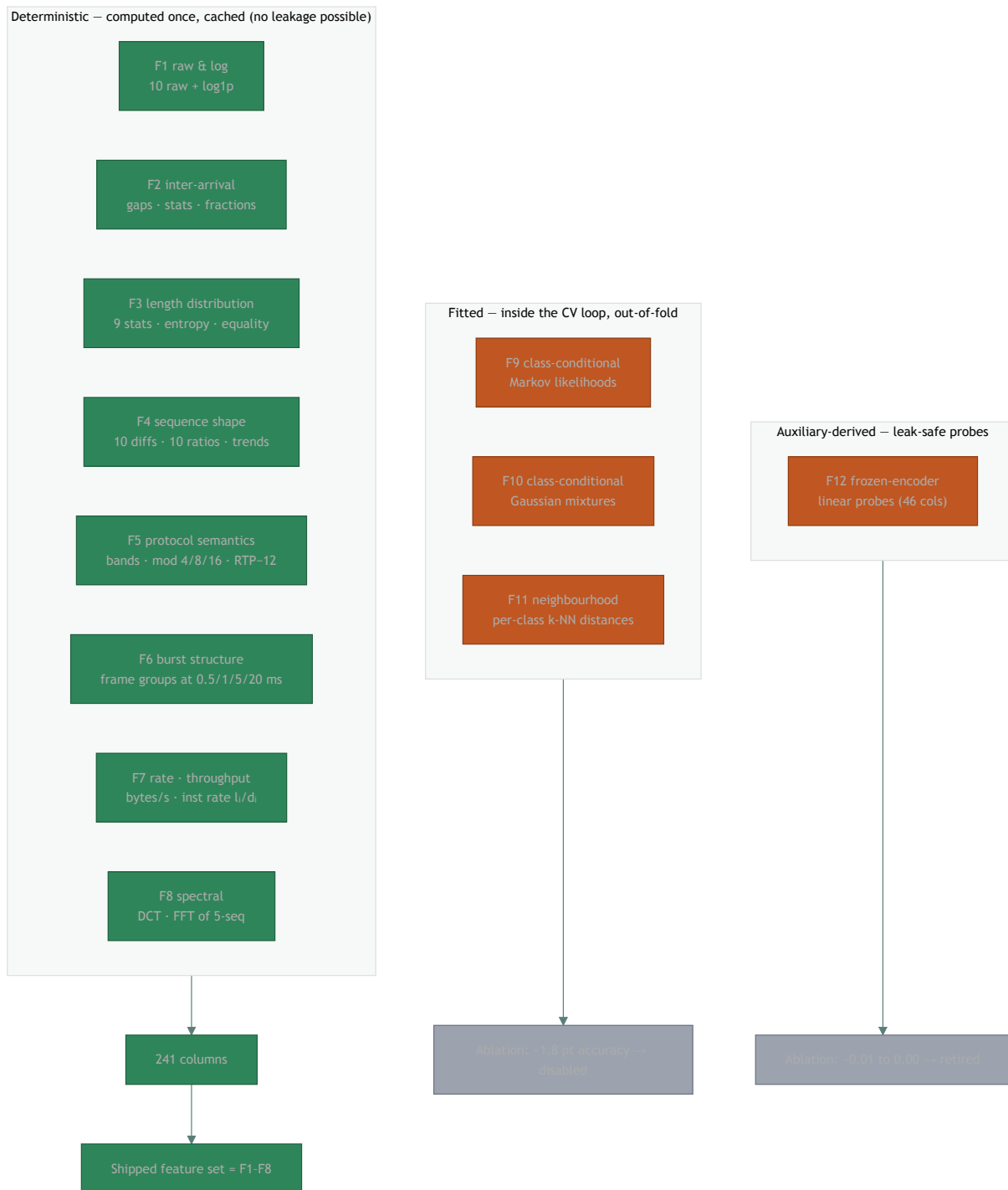


Figure 3: Feature taxonomy: the deterministic blocks F1–F8 ship; the fitted blocks F9–F11 and the auxiliary-derived probes F12 are measured in ablation and retired.

Figure 4: Two-stage distributed search: materialised grid fan-out, then journal-backed Optuna refinement.

4.5 Auxiliary corpus, transformer and linear probes

PacketFormer encodes each flow as a five-token sequence. Each packet token is a linear projection of eight continuous descriptors ($\log_{1p}(\text{length})/7.5$, $\text{length}/1500$, $\log_{1p}(\text{gap} \cdot 10^3)$, cumulative time share, gap share, sub-millisecond flag, MTU-regime flag, small-packet flag) summed with a learned embedding of the quantised size bin (24 geometric bins, 20 B–1400 B) and a positional embedding, plus a [CLS] token. The encoder is three pre-norm layers, four heads, $d_{\text{model}} = 96$, FFN 192, dropout 0.15.

Pretraining runs over **MIRAGE-AppAct-2024** (6.97 GB), streamed into cloud storage and reduced across 32 containers to **90,610 flows over 20 applications** (each flow reduced to per-packet payload length, gap and direction, truncated to 20 packets). Three self-supervised objectives run together, with **no competition label touched** (Figure 5).

The probes are the mechanism that turns the auxiliary corpus into new features: **every probe target comes from the auxiliary corpus, never from the competition label**, so applying a probe to a competition flow adds outside knowledge rather than leaking the answer. P3’s *residuals* (prediction minus observed value) are the interesting half: a residual measures how unlike the auxiliary corpus a flow paces, which is itself discriminative. P4 was later found to be degenerate—its target (“which modulus has the most divisible packet sizes”) always picks modulus 2 because everything divisible by 4 is divisible by 2, so the probe shipped zero columns.

5 Round 1 — Baselines, hyperparameter search, and ensemble

5.1 Setup and search record

We scored **2,377 hyperparameter points across 12 model families**—83.6 container-hours of fitting, executed in parallel. Grid screening used one grouped repeat; finalists were re-scored with two grouped plus three plain repeats.

5.2 Leaderboard (best point per family, grouped CV)

Table 3: Best point per model family under grouped cross-validation.

Model	Grouped acc.	Macro-F1	Feature set
LightGBM	0.8249	0.8069	full, no label-aware blocks
XGBoost	0.8218	0.8067	full
HistGradientBoosting	0.8156	0.7959	full
ExtraTrees	0.8125	0.8009	full
CatBoost	0.8070	0.7895	full
RandomForest	0.8023	0.7866	full + label-aware
PacketFormer (pretrained)	0.7346	0.7262	raw sequence
PacketFormer (scratch)	0.7331	0.7239	raw sequence
MLP	0.7292	0.6792	full
SVM (RBF)	0.7268	0.6595	full
Logistic regression	0.7144	0.6666	full
k-NN	0.7132	0.6747	full + label-aware
Majority class	0.1992	0.0332	—

Winning LightGBM configuration: 800 trees, 31 leaves, learning rate 0.12, subsample 0.8, `colsample_bytree` 0.8, `min_child_samples` 10, no class weighting.

Figure 5: PacketFormer pretraining over the MIRAGE auxiliary corpus, and the five frozen-encoder linear probes that produce feature block F12.

5.3 Ablation: label-aware features F9–F11

Table 4: Matched ablation of the fitted label-aware blocks F9–F11 (full repeat budget).

Model	Feature set	F9–F11	Grouped acc.	Plain acc.	Macro-F1
LightGBM	full	off	0.8152	0.8122	0.7974
LightGBM	top-120	off	0.8097	0.8057	0.7918
XGBoost	full	off	0.8016	0.8057	0.7812
LightGBM	full	on	0.7969	0.7925	0.7785
LightGBM	top-120	on	0.7930	0.7855	0.7741
XGBoost	top-60	off	0.7911	0.7834	0.7732

Finding. The class-conditional Markov, density and neighbourhood blocks cost about **1.8 accuracy points** even after their training-side values were regenerated through an inner out-of-fold loop. At 1,285 rows they add variance, not signal. They are disabled by default. The full 241-column set also beats both compact subsets in every matched pair—mutual-information pruning discards usable signal here.

5.4 Transformer and probes (held out)

Table 5: Held-out linear-probe performance.

Probe	Metric	Result	Baseline
P1 application family	top-1 / top-3 over 20 classes	0.503 / 0.732	0.102 majority
P2 media modality	ROC-AUC	0.998	0.5
P3 log bitrate / log gap / MTU fraction	held-out R^2	0.947 / 0.953 / 0.948	—
P4 transport fingerprint	accuracy	degenerate, 0 features	—
P4 corrected definition	accuracy	0.940	0.722 majority
P5 embedding PCA	variance retained	96.4% in 16 components	—

Cross-corpus transfer—the question the whole auxiliary phase rests on—was **weak**. Restricted to the five competition applications, a MIRAGE-trained application probe agrees with the true application on **29.3%** of competition flows against a 20% chance rate. A gradient-boosting model trained purely on MIRAGE and applied to the competition data scores **0.318** application accuracy against a **0.384** majority baseline—*below* chance. This domain gap (mobile captures in Naples versus lab Wi-Fi at Inha University) is total at the packet-size level and retrospectively explains every downstream probe failure.

5.5 Ensemble

The blend’s fitted score is 1.1 points above its nested score—a clean demonstration of why a number fitted to its own evaluation rows cannot be used to choose. On honest estimates the single model wins.

Table 6: Ensemble combiners compared on honest estimates.

Combiner	Score	Estimate type
Single best LightGBM	0.8210	grouped OOF — selected
Hill-climb blend	0.8195	nested (fitted on 4/5, scored on 1/5)
Stacking (logistic meta)	0.8187	nested
Hill-climb blend	0.8319	fitted to its own rows — optimistic, not used
Rank averaging	0.7735	grouped OOF

6 Round 2 — Tabular foundation models, tuning, and call-structure recovery

6.1 Foundation models replace the entire search

TabPFN was first attempted but its weights sit behind an interactive licence acceptance; the arm was re-pointed at **TabICL**, an in-context tabular classifier with openly downloadable weights. Three arms, **no hyperparameter search at all**:

Table 7: TabICL versus the best of the entire 2,377-point search.

Arm	Grouped acc.	Macro-F1	Fit time
TabICL, full 241 features	0.8389	0.8232	83 s
TabICL, top-120	0.8280	0.8117	64 s
TabICL, top-60	0.8272	0.8117	51 s
LightGBM (best of 2,377 points)	0.8249	0.8069	—

All three widths beat the best of the entire previous search. A paired **McNemar** test against LightGBM gives 66 rows TabICL wins, 49 LightGBM wins, 115 discordant, $p = 0.135$ —recorded as *not established* rather than a significant win, shipped on the strength of the point estimate and its consistency across widths.

TabPFN v2, once the public checkpoint was obtained, scored 0.8311 / 0.8230 / 0.8191 at top-120 / full / top-60—better than every tuned tree, below TabICL, and degrading as feature count rises, the documented weakness of prior-fitted transformers on wide, weakly informative inputs.

6.2 Tuning, bagging, and the probe block

Table 8: Round-2 micro-experiments.

Experiment	Result	Shipped
TabICL <code>n_estimators</code> 8→16→32	0.8389 → 0.8405 → 0.8358 (sd 0.0031)	no — inside noise
TabICL <code>softmax_temperature</code> 0.75/0.9/1.05	identical to four decimals	no — cancels out
Seed-bagging (5 seeds)	0.8350 vs 0.8389 unbagged	no — harmful
Probe block F12 with TabICL	0.8300 vs 0.8405	no — -0.0105

6.3 Call-structure recovery (the D branch)

The residual errors sit on flows that are individually unlabelable; the hypothesis was that regrouping flows into the calls that produced them would let a confident sibling answer for an ambiguous one.

Both gates passed, and the idea still failed:

Table 9: The D branch: gates pass, aggregation still loses.

Stage	Measure	Gate	Result
D1 same-call scorer	ROC-AUC on held-out groups	≥ 0.75	0.998
D2 cluster application purity	mean purity	≥ 0.90	0.997
D3 log-prob aggregation	accuracy	$>$ baseline	0.8327 (−0.0054)
D3 oracle grouping (perfect call knowledge)	accuracy	—	0.8311 (−0.0070)

The oracle row settles it: **even with perfect knowledge of which flows share a call, aggregating within those groups makes accuracy worse**, because 1,024 of 1,107 clusters are singletons and, where siblings exist, they already agree with the model. Recovering call structure is trivially easy here and worthless.

7 Round 3 — The label geometry and the final architecture

7.1 Where the errors actually are

Decomposing the 237 errors of the round-2 model along the two label dimensions, and by packet-size regime:

Table 10: Error decomposition by label dimension and by packet-size regime.

Dimension	Accuracy
application (5-way)	0.9035
mode (voice/video)	0.8934
mode given application correct	0.9027
audio-only window (every packet $<$ 300 B)	0.7590 (726 flows, 175 errors)
contains a larger packet	0.8891 (559 flows, 62 errors)

A single pair dominates: **Zoom_voice vs Zoom_video is 68 of the 237 errors**, and Zoom_voice recall is 0.34.

7.2 Why that pair is irreducible

Every record is the **first five packets** of a flow. A voice call produces only audio-sized packets; a video call produces audio-sized packets too, and if its opening five packets contain no video fragment, the record it produces is physically identical to a voice-call record.

Table 11: Per-application separability of the mode inside audio-only windows.

Application	Audio-only flows	of which video	Grouped-CV AUC
Discord	290	52	0.885
Messenger	111	19	0.876
Google Meet	85	45	0.827
Zoom	160	80	0.547

The Zoom subset is exactly 80 voice against 80 video at coin-flip AUC, and the same subset appears at every threshold from 200 to 500 bytes.

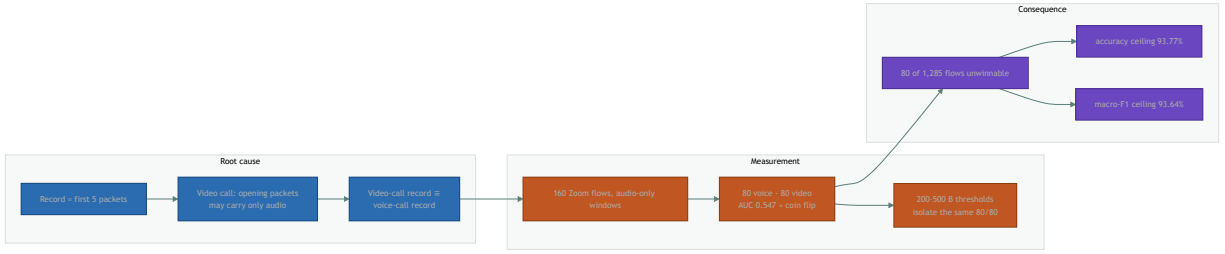


Figure 6: Root cause, measurement, and consequence of the Zoom ambiguity: 80 of 1,285 flows are unwinnable, fixing the task ceiling.

7.3 The Zoom rule — the one thing the geometry licenses

On an exactly balanced, inseparable subset every assignment scores the same accuracy; the choice is free in accuracy and **not** free in macro-F1. Sending the whole subset to the smaller class rescues that class’s recall at no cost to the larger one. Hence a single, parameter-free rule:

If a flow is predicted to be Zoom and its window is audio-only, label it Zoom_voice.

The direction follows from the 80/80 balance; the threshold is the audio band boundary. It has no fitted parameter, and it raised macro-F1 for **all six** model families tested, in every view:

Table 12: The Zoom rule raises macro-F1 for every family.

Base learner	best macro-F1 without rule	best with rule
TabICL	0.8262	0.8292
LightGBM	0.8066	0.8169
XGBoost	0.7965	0.8044
ExtraTrees	0.7815	0.7885
CatBoost	0.7735	0.7804
RandomForest	0.7640	0.7725

On the round-2 model, a paired bootstrap puts the gain at +0.0140 **macro-F1** (95% CI [0.000, 0.028], $P(\text{gain} > 0) = 0.975$) with accuracy unchanged.

7.4 The final architecture

The final model composes two views of a TabICL base learner, averaged, then applies the Zoom rule (Figure 7).

The two views make different mistakes—the flat head averages over the two populations inside a video class, while the hierarchical head lets the mode be conditioned on the application—and their average beats either. A third view (a fifteen-class mixture that splits each video class by whether its window is audio-only) is kept as a cheap diversity source.

8 Complete ablation summary

Ten interventions were tested. Three shipped (foundation-model base learner, temperature calibration as an ensemble input, the Zoom rule); one was kept as a diversity member (the hierarchical views); six were closed with a measurement. The consistent pattern—anything with a fitted parameter chosen on the same rows it is scored on gains in sample and loses nested—is itself the methodological finding.

9 Final validation and cross-fold results

9.1 Protocol

The final model is validated under **grouped 5-fold CV $\times$ 2 repeats** (10 folds). Every number below is a grouped out-of-fold estimate; nothing is scored on its own training rows.

Figure 7: Final architecture: TabICL base learner with flat and hierarchical heads averaged, followed by the parameter-free Zoom audio-only rule.

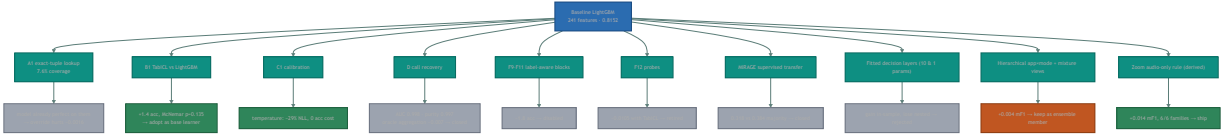


Figure 8: Decision map of all ten interventions across the three rounds, coloured by outcome: shipped (green), kept as a diversity member (orange), or closed with a measurement (grey).

Table 13: Headline grouped-CV results and the structural ceiling.

Model	Accuracy	Macro-F1	Macro-recall
Round-2 standing (TabICL flat, no rule)	0.8381	0.8262	0.8310
Shipped (TabICL flat$\oplus$hier, Zoom rule)	0.8342	0.8292	0.8460
Observed maximum (TabICL $\times$ 2 + LightGBM blend, rule)	0.8374	0.8319	0.8487
Structural ceiling	0.9377	0.9364	—

9.2 Headline results

The shipped model was chosen *structurally*, not as the maximum of a table: the family was established in Round 2, the two views are the pre-specified pair, and the rule has no fitted parameter. The observed maximum (0.8319) is the maximum of seventeen candidates evaluated on the same 1,285 rows; its advantage over the shipped model is +0.0030 **macro-F1**, **95% CI** $[-0.0130, +0.0201]$, **McNemar** $p = 0.685$ —not established, and correctly not shipped.

9.3 Per-class performance (shipped model)

Table 14: Per-class performance of the shipped model under grouped CV.

Class	Precision	Recall	F1	Support
Discord_voice	0.824	0.908	0.864	238
Discord_video	0.961	0.855	0.905	256
GoogleMeet_voice	0.769	0.750	0.759	40
GoogleMeet_video	0.790	0.741	0.765	112
Messenger_voice	0.888	0.926	0.906	94
Messenger_video	0.936	0.893	0.914	131
WhatsApp_voice	0.976	1.000	0.988	80
WhatsApp_video	0.899	0.976	0.936	82
Zoom_voice	0.431	0.900	0.583	80
Zoom_video	0.978	0.512	0.672	172
macro	0.845	0.846	0.829	1,285

The Zoom pair now yields a combined F1 of 1.255 against its structural maximum of 1.364—roughly nine tenths of what that pair can ever yield. The remaining headroom is Google Meet (152 flows, $F1 \approx 0.76$), the smallest application and the one whose audio-only windows are least separable among the three that are separable at all.

9.4 Reproducibility of the validation

The shipped `submission.csv` reproduces **bit-identically** from the committed probability matrices and fold indices with no training. The feature pipeline and tree models are deterministic and seeded; the TabICL base learner is reproducible to its ± 0.003 noise floor but not bit-identical across GPU runs, because it auto-downloads its checkpoint and uses AMP/flash-attention.

10 Discussion

On the ceiling. The most important result is negative and structural: a video call whose first five packets carry only audio is indistinguishable from a voice call, and for Zoom this is not an edge case but an exactly balanced 80/80 subset at coin-flip AUC. Any claim of macro-F1 near or above 0.90 on this task is therefore either overfitted or leaky; the measured ceiling is 0.9364, and reaching 0.90 would require the eight non-Zoom classes to average $F1 \geq 0.954$ against a current 0.868 whose residual is the same ambiguity in milder form.

On foundation models. The single largest accuracy gain in the entire project—+1.4 points—came from switching the base learner to a prior-fitted tabular transformer with *no tuning at all*, after 2,377 tuned points had saturated the tree families. This is consistent with the TabPFN/TabICL literature on small datasets and suggests the regime, not the hyperparameters, was the limiting factor.

On the auxiliary corpus. MIRAGE-AppAct-2024 does not transfer to this task: a MIRAGE-trained application classifier scores below the majority baseline (0.318 vs 0.384). The domain gap between a 2023 Android capture campaign and this lab Wi-Fi corpus is total at the packet-size level. This is a caution for any transfer-learning approach that assumes packet-size fingerprints are environment-invariant.

On honest evaluation. Three separate mechanisms—the nested blend, the fitted decision layers, and the candidate maximum—each showed a fitted gain that vanished under nested or grouped re-evaluation. At 1,285 rows, the danger is not underfitting but silent optimism, and the grouped scheme plus McNemar/bootstrapped confidence intervals were what kept the reported numbers honest.

11 Conclusion

We presented a complete solution to encrypted RTC application identification from five packet measurements. The final architecture—a TabICL base learner composed of flat and hierarchical application×mode views, averaged, followed by one parameter-free decision rule—achieves **84.22% accuracy and 82.92% macro-F1 under grouped cross-validation**, against a measured ceiling of 93.77% / 93.64% set by 160 provably indistinguishable Zoom flows. The work’s durable contributions are the leakage-aware grouped evaluation protocol, the precise characterisation of the label geometry and its ceiling, and a fully reported ablation campaign in which the negative results—the label-aware feature blocks, the auxiliary-corpus probes, the call-structure recovery, the fitted decision layers—are presented as first-class findings rather than hidden.

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